Avery Briggs | abriggs914@gmail.com

# Professional Summary

Software developer and automation engineer with a Bachelor of Computer Science and several years of experience designing data-driven applications, automation tools, and API-driven systems. Experienced in building automated workflows, integrating external APIs, and developing data processing pipelines using Python and SQL. Strong background in debugging complex systems, validating data pipelines, and building tools that automate operational processes across multiple departments. Comfortable working with modern automation frameworks and continuously expanding expertise in Java-based testing environments, CI/CD pipelines, and cloud-hosted services.

# Experience

## Software Systems Developer | Process Analyst

### BWS Manufacturing Ltd. | Centreville, N.B. (Jan 2021 – Feb 2026)

Developed internal software systems and software development tools supporting operational workflows across Sales, Procurement, Production, and Scheduling within a small technical team in a manufacturing environment.

**Key Contributions**

- Full-stack development of Python-based internal applications and Streamlit dashboards used by multiple departments to analyze operational data and automate workflows and support decision-making.
- Developed complex SQL Server data pipelines and analytics queries enabling cross-department analysis of inventory, production, purchasing, and sales metrics.
- Built automation tools that eliminated manual operational tasks, including systems that automatically aggregate and prepare files for reporting and meetings, reducing manual preparation time from hours to seconds.
- Designed and implemented a warehouse visualization system mapping inventory locations onto spatial layouts to improve operational visibility and planning.
- Developed tools using OCR techniques to extract structured data from legacy PDF documents, enabling automated reporting and analysis of previously inaccessible operational records.
- Refactored legacy Microsoft Access applications by consolidating duplicated VBA logic into reusable shared modules, eliminating 3–5 smaller databases and improving maintainability of the primary operational system.
- Built an administrative time-tracking interface integrating live ShopClock data, allowing supervisors to review, adjust, and approve employee shift records before payroll submission, improving visibility and coordination across multiple departments.
- Created and maintained reusable Python utility libraries supporting database access, file management, reporting, and UI helpers used across internal projects.
- Provided technical troubleshooting and support across hardware systems, network devices, ERP integrations, and vendor-managed services.

**Notable Projects:**

- IT Request and Inventory Ticketing System. Built from scratch using SQL and Access.
- Custom drag-n-drop production schedule. CustomTkinter application packaged into an executable file.
- Streamlit dashboards for sales margins and inventory analysis including inputs and reporting.
- 3D warehouse visualization using Plotly, demonstrating inventory bottlenecks and improving transparency.
- PDF parsing of legacy reports. Performed using Python and modules: Pdfplumber, Streamlit, and Pytesseract.
- Implementation of Access version control protocols to maintain program integrity.
- Yellow Tag system to automatically flag, record, and report missed parts when picked for production.

**Technologies used:**

- Python, SQL Server, Streamlit, Pandas, Plotly, Pytesseract, Reportlab, CustomTkinter, Microsoft Access, VBA, Syspro8 ERP, ShopClock, NeWare Security

**Other Job Experience:**

- Basketball Referee – CBOC – Fredericton, NB – Oct 2013 – Mar 2020
- Kitchen Staff – FMI – Fredericton, NB, -- May 2015 – Sept 2017
- Farm Hand – Hunter Brother’s Farm Market – Florenceville, NB, -- Jun 2011 – Aug 2013

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# Education

## Bachelor of Computer Science

### UNB – Fredericton N.B. (May 2020)

- Specialization: Software Systems.
- Recipient of Governor Thomas Carleton Entry Scholarship.

# Skills

- **Programming**: Python, SQL, VBA, Java
- **Frameworks / Libraries**: Streamlit, Pandas, Plotly, Tkinter, CustomTkinter, Pdfplumber, Pytesseract
- **Learning**: TypeScript, C#, .NET, other modern web frameworks
- **Databases**: Microsoft SQL Server, Microsoft Access
- **APIs**: NHL Stats API, IMDB API, Fitbit API
- **Tools:** Git, Syspro ERP, ShopClock
- **Languages**: English (native), French (Intermediate oral proficiency)
- **Other**: Strong cross-team communication and collaboration. Self-directed problem solving and rapid learning.

# Awards & Activities

- Finalist (Top 12) - Kaspersky Academy Partner's Summit (KAPS) Lisbon, Portugal (December 2019).
- Certified intermediate plus in the French oral proficiency exam (June 2013).
- UNB dormitory Orientation Committee (September 2014) & Social Representative (September 2014 – April 2015).
- UNB Relay for Life steering committee logistics team (September 2013 – February 2014).

# Hobbies

3D printing and hardware projects, sports analytics software, recreational athletics (cycling, kayaking, basketball officiating)

# Projects

- Live NHL Scoreboard showcasing game scores and standings data, with an interactive playoff bracket selection. Leverages NHL Stats API to show real-time figures, summarized
- STL file generation for automating the creation of strings of text using custom fonts and other parameters. Uses a Solid2 to generate files, and Streamlit interface to model and tweak them.
- Minesweeper Android mobile application, built using Java and Android Studio.
- Image editing Android application taking advantage of the Gestures library to offer a unique interface for tuning photos.
- Traffic simulation engine using Pygame. Mimics some elements from Cities Skylines traffic patterns.
- COVID-19 interaction simulator built with JavaFX modeling contagion spread in configurable populations.

References available upon request